

Report
Of
Full Stack Developer Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

ROLL NUMBER: 2410031627

NAME: SIMMI TIWARI

BATCH: 2026-27

CANDIDATE'S DECLARATION

I, SIMMI TIWARI, hereby declare that the internship report titled “Full Stack Development Internship and School Management System” has been completed by me in fulfilment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The report presents the work, learning, technical activities, and project experience documented during my internship at Codec Technologies.

I further declare that the references and source materials used in preparing this report have been appropriately acknowledged. The work presented in this report is prepared for academic submission and has not been submitted to any other institute for the award of any degree or diploma. I understand that I shall be solely responsible for any copyright or academic-integrity violation arising from material included in this report.

Signature: _____

Student Name: SIMMI TIWARI

Roll No.: 2410031627

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported and guided me throughout my internship program at Codec Technologies. The internship provided me with valuable exposure to practical software development and helped me understand how academic concepts can be applied to real-world development tasks.

I am thankful to the mentors, senior developers, and team leads whose guidance, constructive feedback, and technical discussions helped me improve my understanding of PHP, MySQL, authentication, authorization, CRUD operations, responsive interface development, database design, and role-based dashboards. Their support encouraged me to approach development tasks systematically and to strengthen my debugging and problem-solving skills.

I am also grateful to IILM University, Greater Noida, and the School of Computer Science and Engineering for providing the academic framework through which this internship experience could contribute to my B.Tech CSE learning. Finally, I express my heartfelt gratitude to my parents for their constant encouragement, support, and blessings.

Signature: Simmi Tiwari

Student Name: SIMMI TIWARI

Roll No.: 2410031627

INTERNSHIP COMPLETION CERTIFICATE



TABLE OF CONTENTS

Chapter No.	Chapter Name	Page Nos.
Cover Page	—	1
1.	Candidate's Declaration	2
2.	Acknowledgment	3
3.	Internship Completion Certificate	4
4.	Project Description	7
4.1	Introduction	7
4.2	Organization Profile	8
4.3	Problem Statement	9
4.4	Project Objectives	9
4.5	Scope of the Project	10
4.6	Technologies and Tools Used	11
4.7	System Architecture	12
4.8	Methodology	13
4.8.1	Development Workflow	13
4.8.2	Functional Modules	14
4.8.3	Weekly Logbook	15
4.8.4	Challenges and Solutions	16
4.8.5	Technical and Soft Skills Learned	16
4.8.6	Proof of Work	17
4.9	Expected Outcomes	23
4.10	Certificates and Communication Proof	23
5.	Bibliography/References	24

LIST OF TABLES

Table No.	Title	Page No.
Table 4.1	Internship Profile	7
Table 4.2	Technologies and Tools Used	11
Table 4.3	Role-Based Access Structure	13
Table 4.4	Functional Modules of the School Management System	14
Table 4.5	Six-Week Internship Logbook	15
Table 4.6	Challenges and Solutions	16
Table 4.7	Technical and Soft Skills Learned	16

LIST OF FIGURES

Figure No.	Title	Page No.
Figure 4.1	Conceptual System Architecture	12
Figure 4.2	Authentication and Role-Based Access Flow	14
Figure 4.3	Login / Register Interface	17
Figure 4.4	Administrator Dashboard	17
Figure 4.5	Fee Management Interface	18
Figure 4.6	Student Dashboard	18
Figure 4.7	Mark Attendance Interface	19
Figure 4.8	Student Timetable	19
Figure 4.9	Teacher Dashboard	20
Figure 4.10	Attendance Interface	20
Figure 4.11	Marks Entry Interface	21
Figure 4.12	Manage Timetables Interface	21
Figure 4.13	My Fees Interface	22
Figure 4.14	Payment / Fee Records Interface	22
Figure 4.15	Welcome / System Entry Interface	23

4. PROJECT DESCRIPTION

4.1 Introduction

The internship at Codec Technologies Pvt. Ltd. provided practical exposure to full-stack web development and helped bridge academic learning with real software development activities. According to the internship certificate, the internship was conducted from 11/05/2026 to 11/07/2026 as a two-month AICTE & ICAC approved internship program in the role of Full Stack Developer Intern.

During the internship, the work documented in the supplied report focused on PHP and MySQL based web development, database design, CRUD operations, authentication and authorization, role-based access control, responsive user-interface design, and GitHub based version control. A major practical outcome was contribution to a School Management System containing role-based dashboards and operational interfaces for administrators, teachers, and students.

The project-oriented nature of the internship allowed the development concepts studied academically to be applied in an integrated environment. Instead of treating frontend, backend, and database concepts as isolated topics, the work connected user interfaces with server-side processing and persistent data storage. This made the internship an important learning experience in understanding the end-to-end structure of a full-stack application.

Table 4.1: Internship Profile

Parameter	Details
Student	SIMMI TIWARI
Roll Number	2410031627
Section	3CSE9
Organization	Codec Technologies Pvt. Ltd.
Internship Role	Full Stack Developer Intern
Internship Duration	11/05/2026 – 11/07/2026
Program	2 Month AICTE & ICAC Approved Internship
University	IILM University, Greater Noida

4.2 Organization Profile

Codec Technologies is presented in the supplied internship report as a digital transformation and software development company working on practical technology solutions for institutions and organizations. The documented areas include management systems, business process automation tools, custom full-stack web applications, and secure dashboards and portals.

The organization's work environment, as reflected in the internship tasks, exposed the intern to the complete flow of web application development. This included designing interfaces, processing forms, implementing server-side logic, managing relational data, applying authentication and role-based authorization, and maintaining development work through GitHub.

The internship therefore provided exposure to both technical implementation and professional development practices. The project environment required attention to functional correctness, database consistency, responsive interface behaviour, access control, debugging, and communication with senior developers and team leads.

The supplied certificate identifies Codec Technologies Pvt. Ltd. as the organization from which the internship certificate was awarded and identifies the internship designation as Full Stack Developer Intern.

4.3 Problem Statement

Educational institutions handle a large amount of information related to students, teachers, attendance, marks, fees, timetables, and announcements. When these activities are handled through disconnected records or separate manual processes, information can become difficult to maintain, retrieve, and update consistently. Different users also require different levels of access to institutional information.

The School Management System addressed in the internship project was designed as a centralized web-based environment in which role-based users could access relevant dashboards and functions. The documented project included administrator, teacher, and student interfaces and provided functionality related to attendance, fee management, marks, timetables, and other school-management operations.

From a software-development perspective, the problem also involved implementing secure access to role-specific functionality, connecting application forms with a relational database, performing CRUD operations reliably, and presenting the resulting information through responsive interfaces.

Therefore, the core problem can be stated as: to develop and support a web-based school management environment that organizes institutional information through a centralized database, provides role-based access, and offers usable interfaces for the different categories of users.

4.4 Project Objectives

- To gain practical experience in full-stack web development using PHP, MySQL, HTML, CSS, Tailwind CSS, and JavaScript.
- To understand the complete flow of a web application from user interface and form submission to server-side processing and database interaction.
- To implement and understand CRUD operations for managing application data.
- To develop authentication and authorization workflows and apply Role-Based Access Control (RBAC).
- To understand relational database design and SQL integration using MySQL and phpMyAdmin.
- To design responsive and user-friendly dashboard interfaces using HTML and Tailwind CSS.
- To understand how role-specific dashboards can provide different functionality to administrators, teachers, and students.

- To practice systematic debugging and troubleshooting of backend logic, database connectivity, and user-interface behaviour.
- To use GitHub as a version-control and development-collaboration tool.
- To gain experience working on a practical School Management System rather than isolated programming exercises.

4.5 Scope of the Project

The scope of the documented project covers the development and practical use of a web-based School Management System with multiple user roles. The system is structured around role-specific dashboards and operational modules, allowing the application to present information and actions according to the authenticated user.

4.5.1 Administrative Scope

- Administrator dashboard and overview of system information.
- Fee-management related interfaces and records.
- Management of timetable information.
- Management of academic and operational data through CRUD-based workflows.
- Role-aware access to administrative functions.

4.5.2 Teacher Scope

- Teacher dashboard for access to relevant teaching functions.
- Attendance-related interface for recording and managing attendance.
- Marks-entry interface for entering academic marks.
- Timetable management and access to teaching schedules.

4.5.3 Student Scope

- Student dashboard providing a personalized overview.
- Viewing attendance information.
- Viewing timetable information.
- Viewing fee-related information through the My Fees interface.
- Access to marks/academic information represented within the supplied project screens.

4.5.4 Technical Scope

Technically, the project scope includes frontend presentation, PHP-based application logic, MySQL database interaction, authentication and sessions, RBAC, CRUD operations, database design through phpMyAdmin, responsive UI development, and

GitHub-based version control. The supplied internship report also documents work with forms, cookies, file uploads, SQL integration, and responsive UI design during the weekly learning process.

4.6 Technologies and Tools Used

The internship report identifies a practical web-development stack in which frontend technologies were used to build interfaces, PHP was used for server-side application logic, and MySQL was used for persistent relational data. Supporting tools such as phpMyAdmin, VS Code, and GitHub formed part of the development workflow.

Table 4.2: Technologies and Tools Used

Category	Technologies / Tools	Application in Internship
Frontend	HTML, CSS, Tailwind CSS, JavaScript	Interface development, responsive layouts, forms and dashboards
Backend	PHP	Server-side processing, forms, sessions, authentication and application logic
Database	MySQL	Persistent relational storage and SQL integration
Database Tool	phpMyAdmin	Database/schema management and inspection
Development IDE	VS Code	Writing, editing and debugging source code
Version Control	GitHub	Source-code version control and project management
Core Concepts	CRUD, Sessions, Authentication, RBAC	Data operations, login state and role-based access

4.6.1 PHP

PHP formed the primary server-side development technology documented in the internship. The weekly learning progression included PHP basics, forms, sessions, cookies, file upload handling, CRUD development, SQL integration, login systems, and RBAC.

4.6.2 MySQL and Database Design

MySQL was used as the relational database technology for storing and retrieving application information. Database schema design and SQL integration were part of the internship learning process, while phpMyAdmin was used as the database-management tool.

4.6.3 HTML, CSS, Tailwind CSS and JavaScript

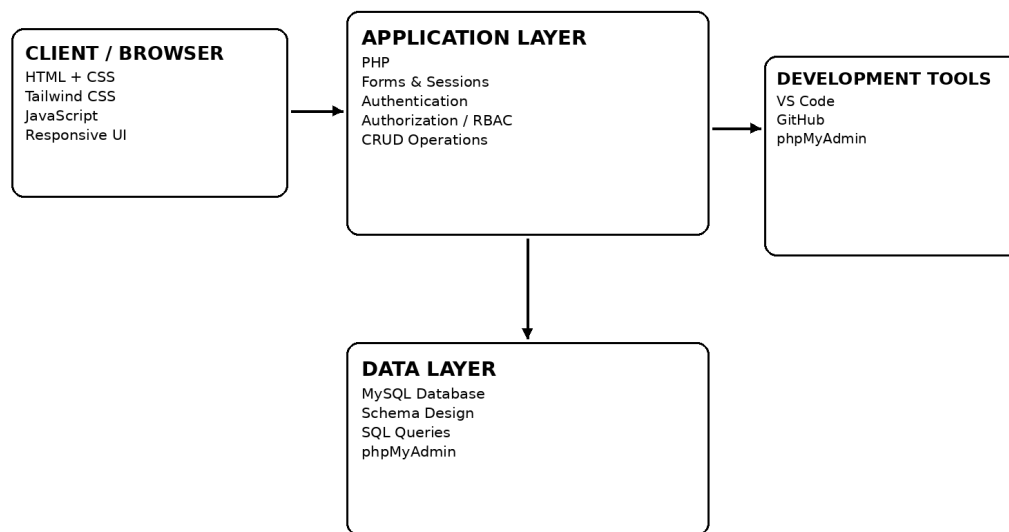
The frontend layer used HTML and CSS concepts together with Tailwind CSS for responsive interfaces. JavaScript was included among the web-development skills learned. The supplied screenshots show dashboard-style interfaces with navigation menus, cards, forms, tables, filters, and role-specific sections.

4.6.4 GitHub and VS Code

VS Code was used as the development environment and GitHub was used for version control. These tools supported organized source-code management and the practical workflow of developing and refining the project.

4.7 System Architecture

The architecture below is a conceptual representation derived from the technologies and workflows explicitly documented in the internship report. It shows how the browser-based interface communicates with the PHP application layer, which in turn interacts with the MySQL database. Authentication, sessions and RBAC operate within the application layer to control access to role-specific dashboards.



4.7.1 Presentation Layer

The presentation layer consists of the browser-based interfaces developed using HTML, CSS, Tailwind CSS and JavaScript. The supplied project screens demonstrate dashboard layouts for administrators, teachers and students, along with forms, tables, navigation menus and status cards.

4.7.2 Application Layer

The application layer is represented by PHP-based processing. It receives form inputs, performs application logic, manages sessions, supports authentication and authorization, and executes CRUD and SQL operations required by the different modules.

4.7.3 Data Layer

The data layer uses MySQL for relational storage. Database schema design and SQL integration were documented as part of the internship. phpMyAdmin was used as the supporting database-management interface.

4.7.4 Role-Based Access

Table 4.3: Role-Based Access Structure

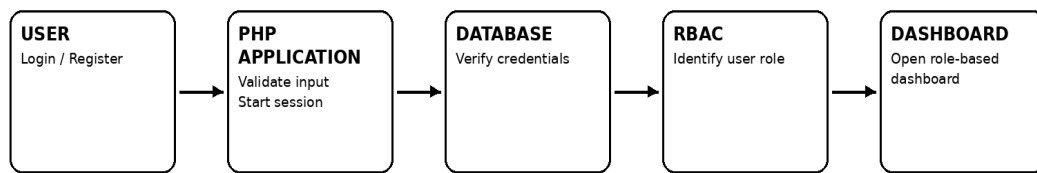
Role	Documented Access / Functions
Administrator / Principal	Administrative dashboard, fee-related management, timetable management and other system-management functions shown in the project.
Teacher	Teacher dashboard, attendance, marks entry and timetable-related functions.
Student	Student dashboard, attendance view, timetable view and fee-related information.

4.8 Methodology

The internship followed a practical, incremental development methodology. The supplied weekly logbook shows a progression from fundamental PHP concepts to sessions and file handling, then CRUD and SQL integration, authentication and RBAC, responsive UI design, and finally development work on the School Management System. This sequence provided a foundation before the concepts were applied to the larger project.

4.8.1 Development Workflow

- Understand the functional requirement or module.
- Design the required form, dashboard, table or interface.
- Implement server-side logic in PHP.
- Connect the required operation to MySQL using SQL integration.
- Apply authentication, session handling and RBAC where required.
- Test the functionality and debug errors.
- Refine the interface for responsive presentation.
- Maintain development work using GitHub.



4.8.2 Functional Modules

Table 4.4: Functional Modules of the School Management System

Module	Documented Function
Authentication Module	Login and registration interface, input handling, sessions and authentication workflow.
Authorization / RBAC	Role-based access to administrator, teacher and student dashboards and functions.
Administrator Dashboard	Dashboard view containing system-level information and access to management functions.
Fee Management	Fee-management interfaces and payment/record information.
Attendance	Attendance viewing and attendance-record management interfaces.
Marks Management	Teacher-facing marks-entry interface and academic marks information.
Timetable Management	Creation/management and viewing of timetable information.
Student Dashboard	Personalized student overview with access to academic and administrative information.
Teacher Dashboard	Teacher-specific overview and access to teaching functions.

4.8.2.1 Authentication and Sessions

Authentication was a major part of the internship learning progression. The weekly logbook explicitly includes sessions and cookies in Week 2 and login system plus RBAC in Week 4. The project screens include login and registration interfaces, demonstrating the user-facing entry point to the system.

4.8.2.2 CRUD and SQL Integration

CRUD operations—Create, Read, Update and Delete—were documented as part of the internship responsibilities. These operations provide the foundation for managing records in a database-backed application. SQL integration connects the PHP

application to MySQL so that records can be created, retrieved, updated and managed through application interfaces.

4.8.2.3 Dashboard Interfaces

The supplied screenshots demonstrate separate dashboards for administrative, teacher and student users. This separation supports the principle that each user should receive information and actions relevant to the assigned role.

4.8.3 Weekly Logbook

The supplied internship report documents a six-week learning and development logbook. Each week is represented by a major technical area and approximately 21–24 hours of work, with the report indicating more than 120 hours completed.

Table 4.5: Six-Week Internship Logbook

Week	Work Summary	Hours
Week 1	PHP Basics, Forms, phpinfo	21–24 hrs
Week 2	Sessions, Cookies, File Upload	21–24 hrs
Week 3	CRUD + SQL Integration	21–24 hrs
Week 4	Login System + RBAC	21–24 hrs
Week 5	UI/UX Responsive Design	21–24 hrs
Week 6	School Management System Development	21–24 hrs

Week 1 – PHP Basics and Forms

The first week focused on PHP fundamentals and form handling. The learning activity created the foundation required for server-side web development and introduced the structure of PHP-based pages and form processing.

Week 2 – Sessions, Cookies and File Upload

The second week introduced mechanisms for maintaining user state and handling additional web-application inputs. Sessions and cookies are particularly relevant to authenticated applications because they support continuity between requests.

Week 3 – CRUD and SQL Integration

The third week focused on CRUD operations and connecting PHP logic with SQL-based database operations. This stage was important for understanding how user actions are translated into persistent database changes.

Week 4 – Login System and RBAC

The fourth week focused on authentication and role-based authorization. This learning directly supported the later role-specific dashboard structure of the School Management System.

Week 5 – Responsive UI/UX Design

The fifth week focused on responsive user-interface design using HTML and Tailwind CSS. The project screenshots reflect dashboard-based layouts designed to organize navigation, cards, forms and tabular information.

Week 6 – School Management System Development

The sixth week focused on development of the School Management System. The supplied report documents role-based dashboards for Principal/Administrator, Teachers and Students and provides screenshots of multiple operational interfaces.

4.8.4 Challenges and Solutions

Table 4.6: Challenges and Solutions

Challenge	Solution Implemented
Difficulties in backend RBAC logic	Consulted the mentor and debugged the logic systematically, testing each step to identify the root cause.
Database connection and query issues	Refined the database schema and added proper error-handling to ensure stable connectivity.

The documented challenges demonstrate that the internship involved more than simply writing new code. Debugging, tracing the source of errors, refining database structures, and discussing problems with mentors were part of the practical learning process.

4.8.5 Technical and Soft Skills Learned

Table 4.7: Technical and Soft Skills Learned

Skill Type	Skills / Learning
Web Development	HTML, CSS, Tailwind CSS, PHP, JavaScript
Database	MySQL, database schema design
Tools	VS Code, phpMyAdmin, GitHub
Core Concepts	Authentication, CRUD, sessions, role-based access
Professional Skills	Communication, collaboration, debugging, analytical thinking

The internship report also documents collaboration with senior developers and team leads, professional communication during discussions and project meetings, improved

analytical thinking through debugging and troubleshooting, and experience working in a real-time project environment.

4.8.6 Proof of Work

The following figures reproduce the project-interface screenshots supplied in the original internship report. They are presented as evidence of the documented School Management System work. Figure numbering follows the college requirement that figures contain chapter-based numbers.

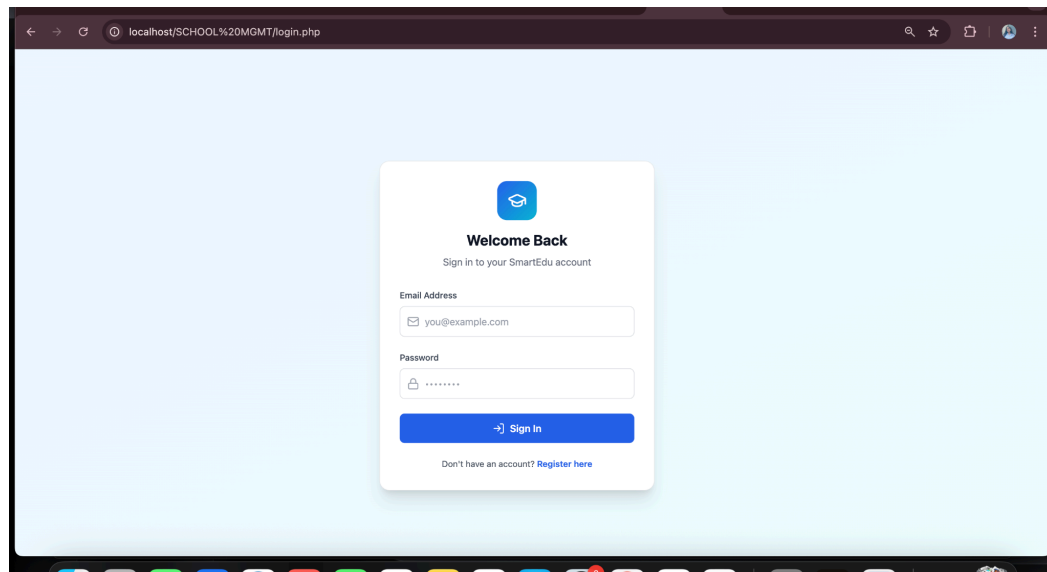


Figure 4.3: Login / Register Interface

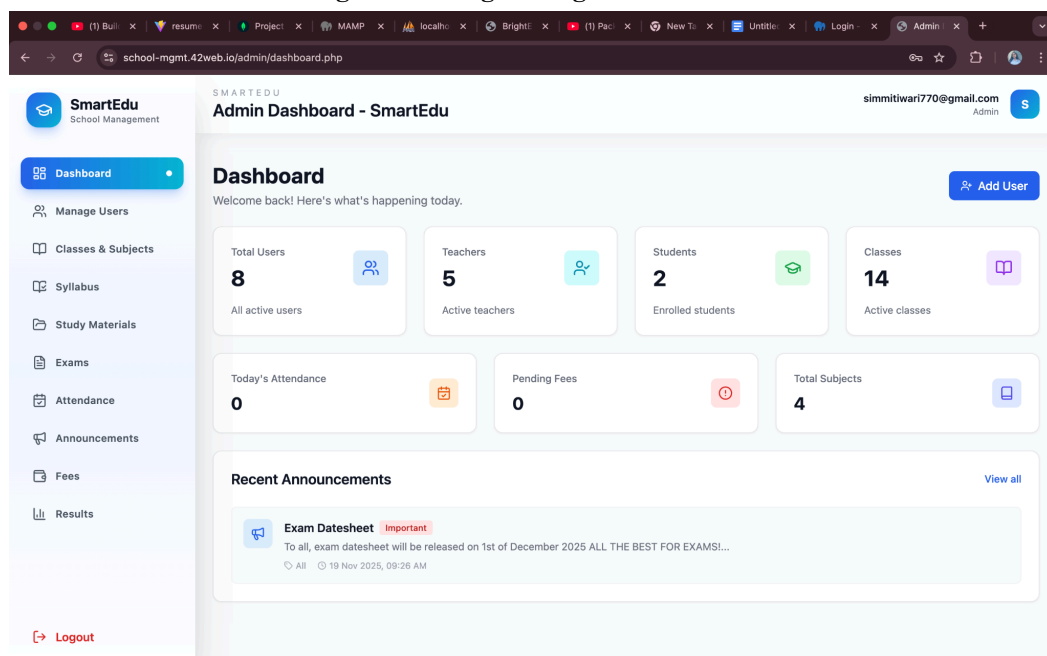


Figure 4.4: Administrator Dashboard

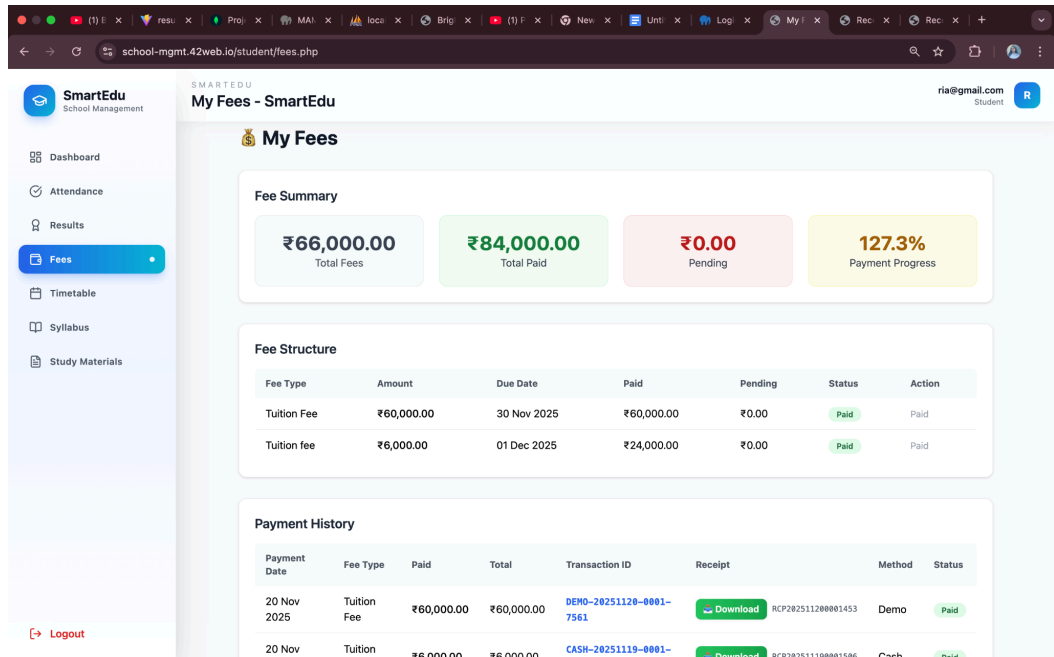


Figure 4.5: Fee Management Interface

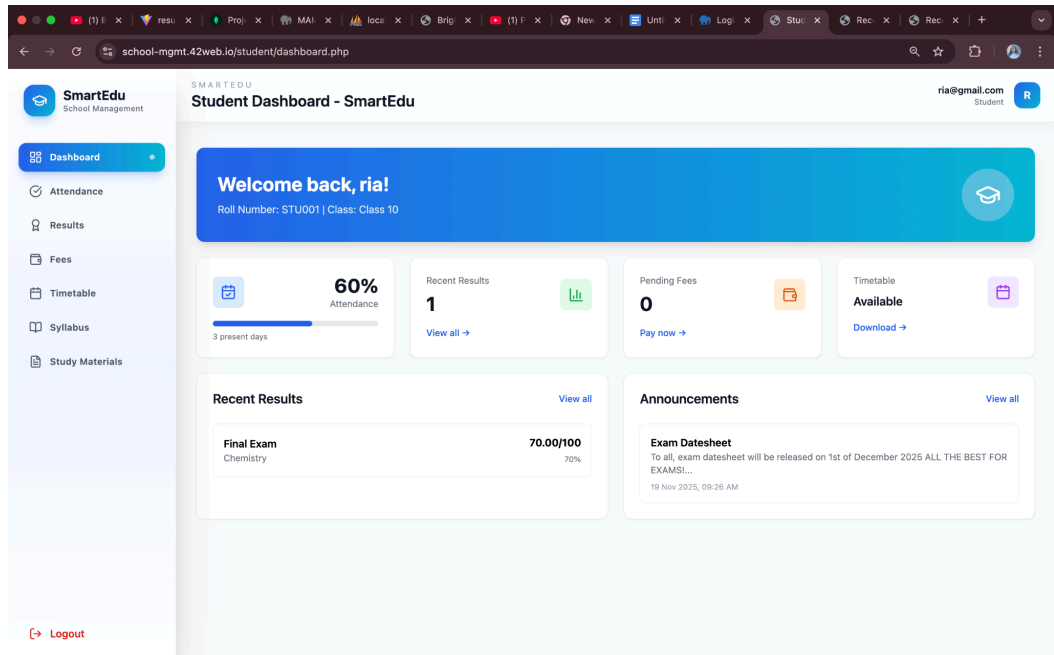


Figure 4.6: Student Dashboard

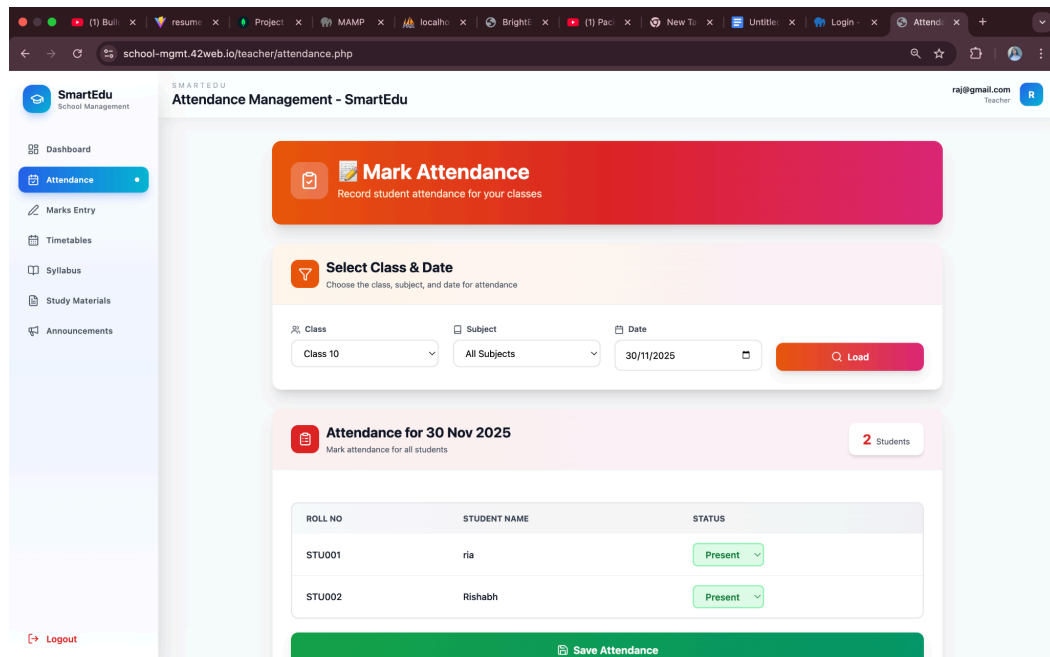


Figure 4.7: Mark Attendance Interface

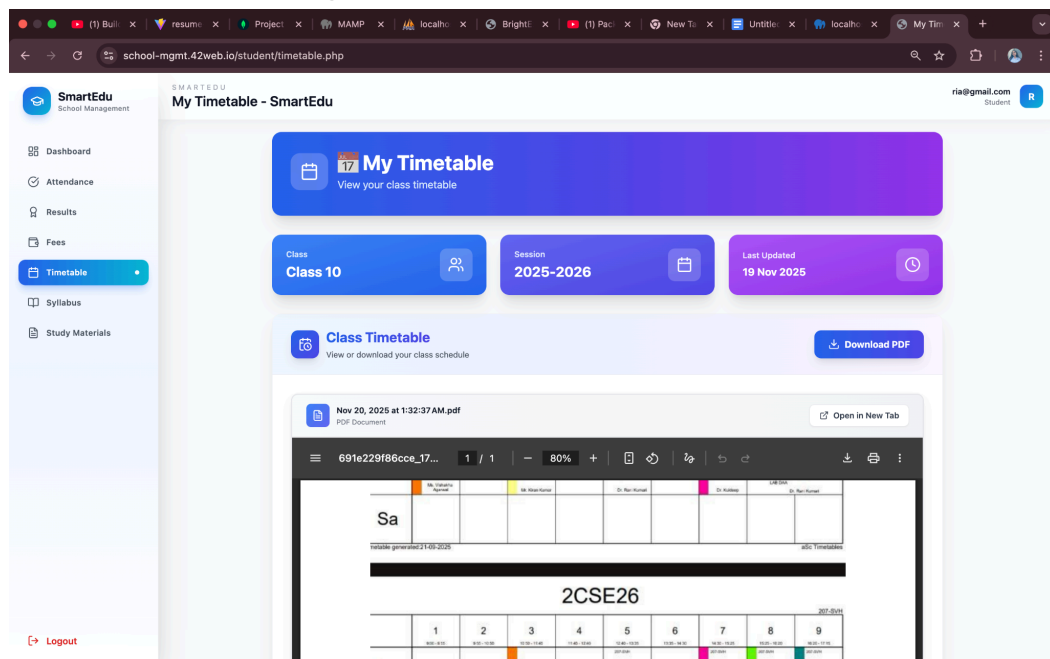


Figure 4.8: Student Timetable

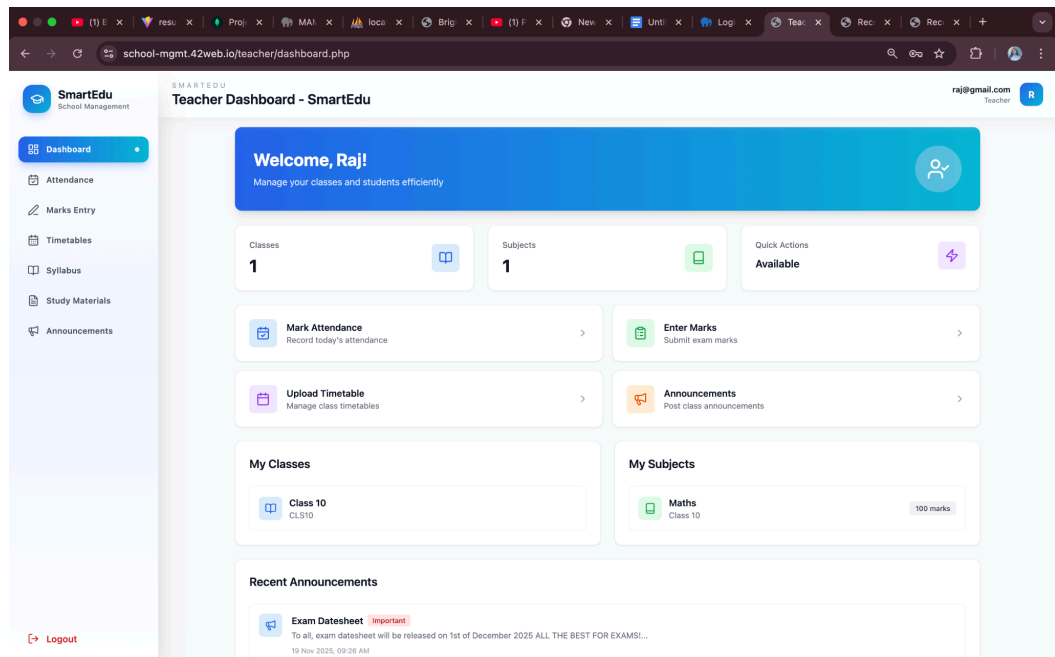


Figure 4.9: Teacher Dashboard

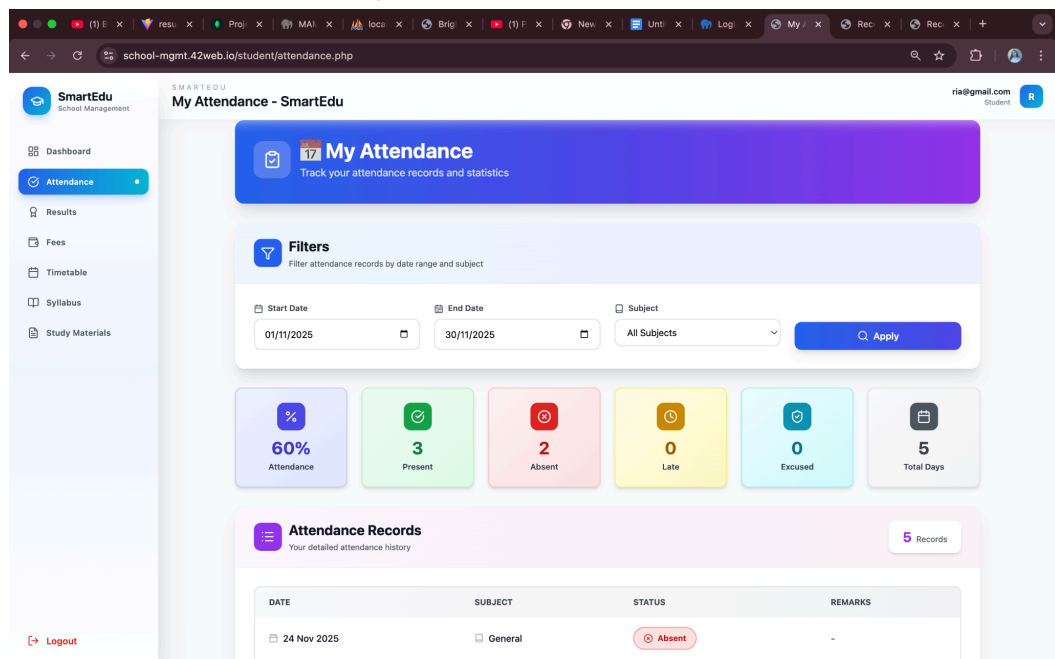


Figure 4.10: Attendance Interface

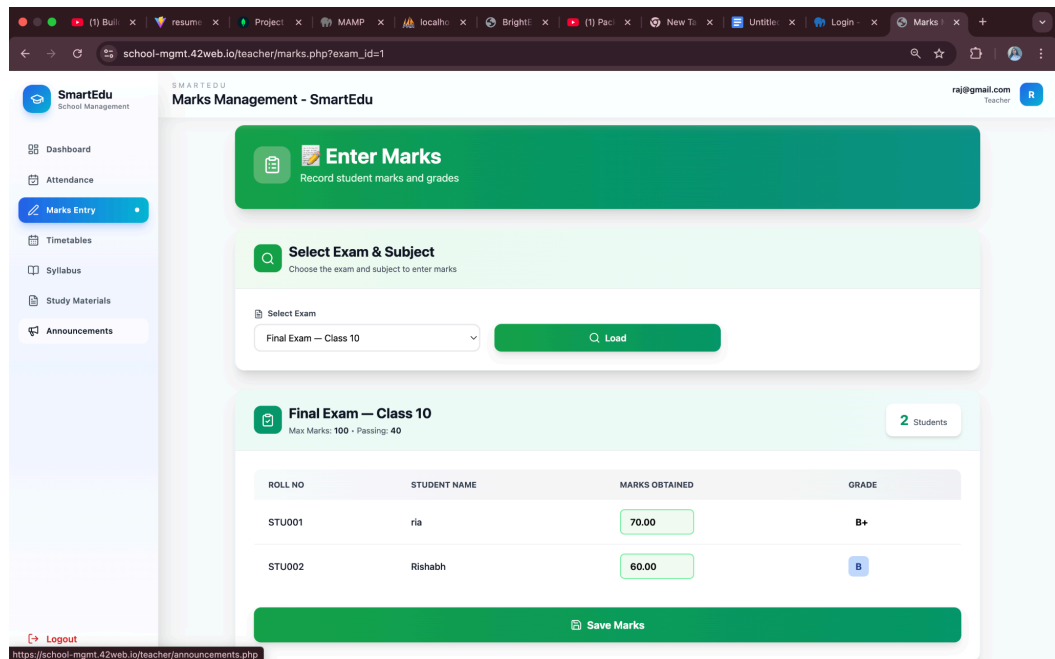


Figure 4.11: Marks Entry Interface

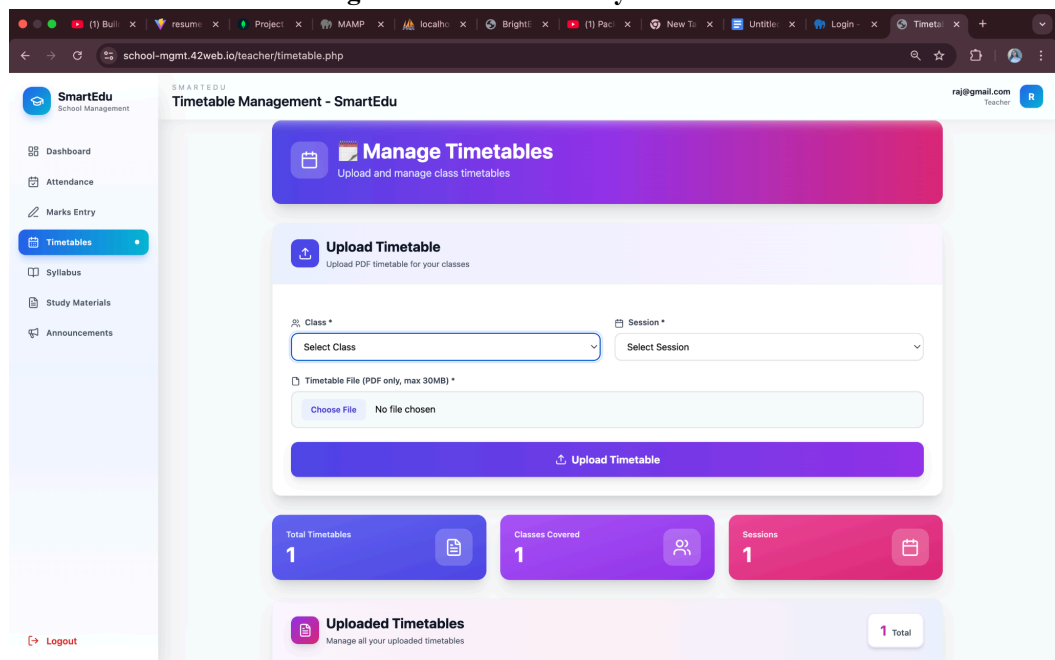
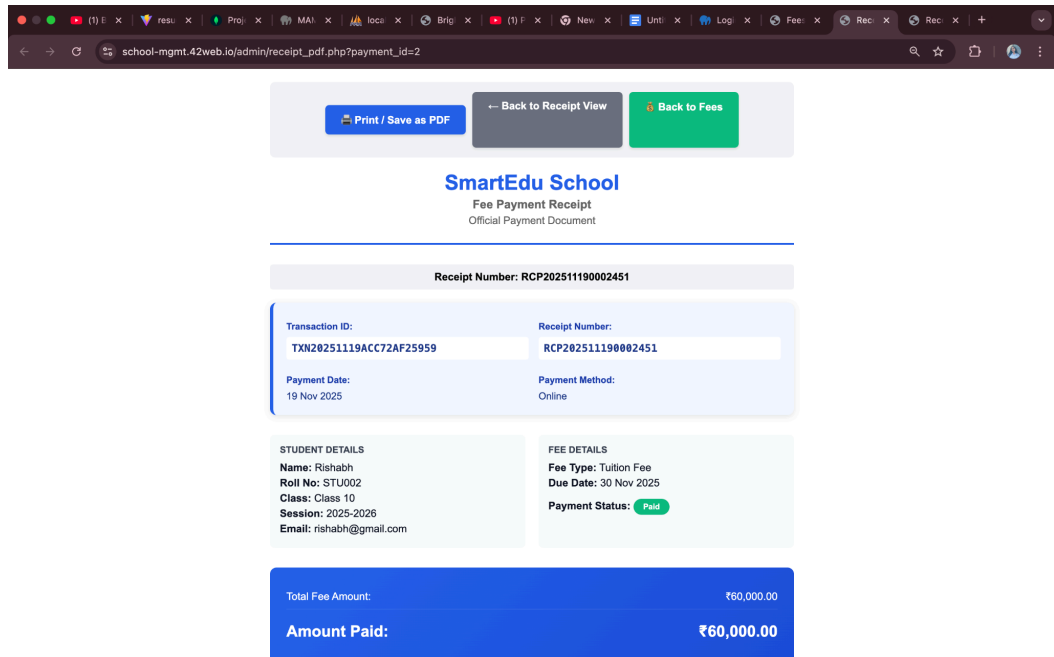


Figure 4.12: Manage Timetables Interface



school-mgmt.42web.io/admin/receipt_pdf.php?payment_id=2

Print / Save as PDF ← Back to Receipt View → Back to Fees

SmartEdu School

Fee Payment Receipt
Official Payment Document

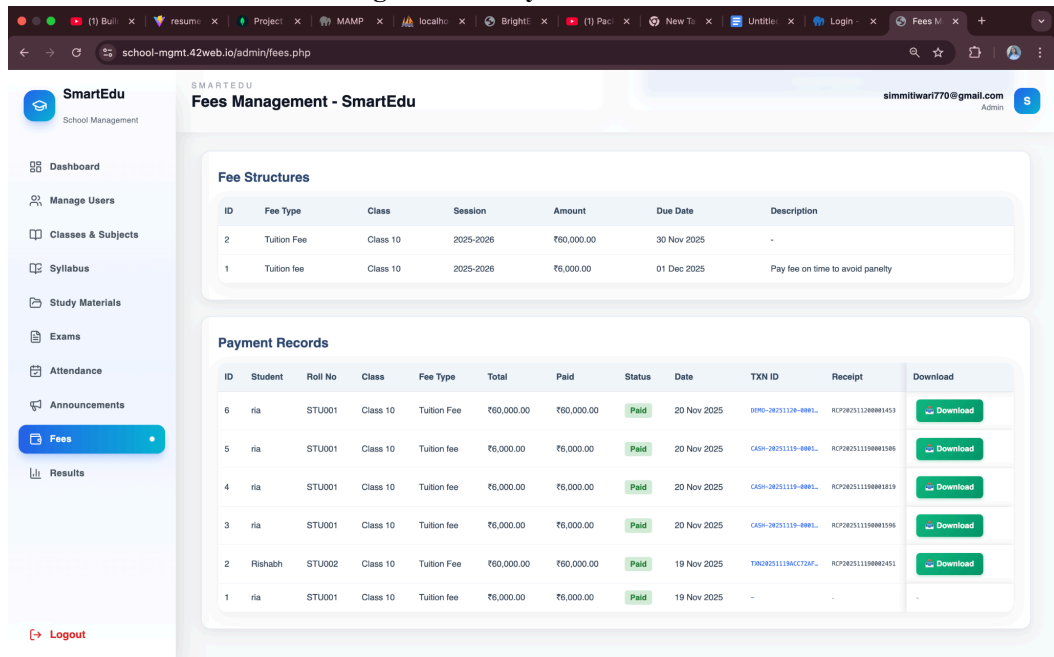
Receipt Number: RCP202511190002451

Transaction ID: TXN20251119ACC72AF25959	Receipt Number: RCP202511190002451
Payment Date: 19 Nov 2025	Payment Method: Online

STUDENT DETAILS Name: Rishabh Roll No: STU002 Class: Class 10 Session: 2025-2026 Email: rishabh@gmail.com	FEE DETAILS Fee Type: Tuition Fee Due Date: 30 Nov 2025 Payment Status: Paid
---	---

Total Fee Amount:	₹60,000.00
Amount Paid:	₹60,000.00

Figure 4.13: My Fees Interface



school-mgmt.42web.io/admin/fees.php

SMARTEDU
Fees Management - SmartEdu

simmitiwar770@gmail.com Admin

SmartEdu School Management

- Dashboard
- Manage Users
- Classes & Subjects
- Syllabus
- Study Materials
- Exams
- Attendance
- Announcements
- Fees**
- Results

Logout

Fee Structures

ID	Fee Type	Class	Session	Amount	Due Date	Description
2	Tuition Fee	Class 10	2025-2026	₹60,000.00	30 Nov 2025	-
1	Tuition fee	Class 10	2025-2026	₹6,000.00	01 Dec 2025	Pay fee on time to avoid penalty

Payment Records

ID	Student	Roll No	Class	Fee Type	Total	Paid	Status	Date	TXN ID	Receipt	Download
6	ria	STU001	Class 10	Tuition Fee	₹60,000.00	₹60,000.00	Paid	20 Nov 2025	BPM-20251120-888L	RCP202511200001453	Download
5	ria	STU001	Class 10	Tuition fee	₹6,000.00	₹6,000.00	Paid	20 Nov 2025	CASH-20251119-888L	RCP202511190001586	Download
4	ria	STU001	Class 10	Tuition fee	₹6,000.00	₹6,000.00	Paid	20 Nov 2025	CASH-20251119-888L	RCP202511190001819	Download
3	ria	STU001	Class 10	Tuition fee	₹6,000.00	₹6,000.00	Paid	20 Nov 2025	CASH-20251119-888L	RCP202511190001596	Download
2	Rishabh	STU002	Class 10	Tuition Fee	₹60,000.00	₹60,000.00	Paid	19 Nov 2025	TXN20251119ACC72AF	RCP202511190002451	Download
1	ria	STU001	Class 10	Tuition fee	₹6,000.00	₹6,000.00	Paid	19 Nov 2025	-	-	-

Figure 4.14: Payment / Fee Records Interface

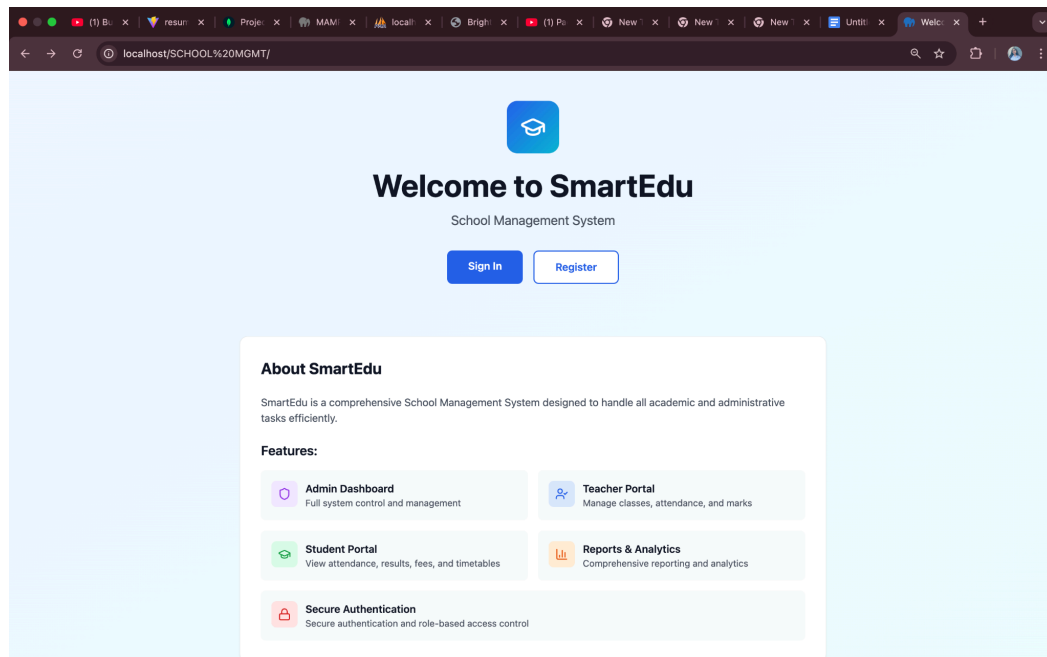


Figure 4.15: Welcome / System Entry Interface

4.9 Expected Outcomes

- Improved practical understanding of full-stack web application development.
- Ability to connect frontend interfaces with PHP-based backend processing and MySQL data storage.
- Practical understanding of CRUD operations and SQL integration.
- Improved understanding of authentication, sessions and role-based authorization.
- Experience designing responsive dashboard-style interfaces using HTML and Tailwind CSS.
- Understanding of database schema design and the use of phpMyAdmin.
- Improved debugging, troubleshooting and analytical problem-solving skills.
- Practical exposure to a School Management System with administrator, teacher and student roles.

The supplied internship report concludes that the experience improved technical development skills and provided real-world experience in building a complete full-stack system. It also records learning in backend logic, UI design, teamwork and development workflow.

4.10 Certificates of Completion and Other Communication Mail Proof

The original internship completion certificate issued by Codec Technologies Pvt. Ltd. is attached in Section 3 of this report. The certificate identifies SIMMI TIWARI as the recipient and records the role as Full Stack Developer Intern for the two-month

AICTE & ICAC approved internship program conducted from 11/05/2026 to 11/07/2026.

The college format also requires other communication mail proof to be attached. No separate communication-mail screenshot or PDF was included in the source files supplied for preparation of this report. Accordingly, the actual communication proof should be inserted by the student immediately after this section before final printing/submission.

5. BIBLIOGRAPHY / REFERENCES

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